YOLOv8-Based Automated Skin Lesion Detection

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July 15, 2026

Abstract

Early detection of skin cancer is critical for successful treatment outcomes, with five-year survival rates exceeding 99% for early-stage melanoma. This paper presents a comprehensive real-time automated skin lesion detection system based on the YOLOv8 deep learning architecture. Unlike traditional classification approaches that provide only image-level predictions, our object detection framework delivers precise spatial localization of lesions within dermoscopic images, better aligning with clinical workflows. The YOLOv8n (nano) variant achieves a mean Average Precision (mAP@0.5) of 93.48% with 93.02% recall and 86.96% precision on dermoscopic images. The high recall rate ensures minimal missed detections (only 7% false negative rate), which is critical for screening applications where the cost of missed lesions far exceeds that of false positives. Furthermore, the lightweight architecture enables real-time inference at approximately 15 milliseconds per image on standard GPU hardware, making it suitable for clinical deployment in resource-constrained environments. Our comprehensive evaluation demonstrates that single-stage object detectors can effectively support dermatological screening applications, offering significant advantages over classification-only approaches by providing actionable spatial information for clinicians while maintaining computational efficiency.

Keywords: Skin lesion detection, YOLOv8, object detection, deep learning, medical imaging, computer-aided diagnosis, dermoscopy, real-time detection, clinical deployment

1 Introduction

1.1 Background and Motivation

Skin cancer represents one of the most prevalent malignancies globally, with melanoma being the deadliest form among various types of skin cancers. According to the World Health Organization, one in three cancers diagnosed worldwide is a skin cancer, with incidence rates continuing to rise annually across most demographics [1]. The increasing prevalence is attributed to multiple factors including aging populations, increased ultraviolet radiation exposure, and improved detection methods leading to higher reported incidence.

Early detection significantly improves prognosis and treatment outcomes. Clinical data demonstrates that five-year survival rates exceed 99% for early-stage melanoma when detected and treated promptly [2]. However, as the disease progresses to advanced stages, survival rates decline dramatically, emphasizing the critical importance of early screening and detection programs.

Traditional visual examination by dermatologists, while effective, faces several significant challenges. First, there exists substantial inter-observer variability in lesion assessment, with studies showing agreement rates varying between 60-80% even among experienced dermatologists. Second, there is limited specialist access in rural and underserved areas, creating healthcare disparities. Third, increasing patient volumes and screening demands strain healthcare systems globally. Fourth, the subjective nature of visual assessment can lead to both false positives (unnecessary biopsies) and false negatives (missed diagnoses).

Computer-aided diagnosis (CAD) systems leveraging advances in deep learning have shown tremendous promise in assisting dermatologists with lesion assessment and diagnosis. These systems can provide objective, consistent evaluations and serve as decision support tools. Most existing research work focuses primarily on image-level classification [3], determining whether an entire image contains a lesion and potentially classifying its type. While valuable, such approaches provide no spatial information about lesion location, limiting their clinical utility.

Object detection frameworks, which simultaneously classify and precisely localize lesions within images, better align with clinical workflows by highlighting exact lesion locations for physician review. These systems can process whole-body imaging or images containing multiple lesions, providing bounding boxes and confidence scores that guide clinical attention and decision-making.

The YOLO (You Only Look Once) family of single-stage detectors offers real-time capabilities crucial for clinical deployment where processing speed directly impacts workflow integration. YOLOv8, released by Ultralytics in 2023, introduces significant architectural improvements over previous versions including anchor-free detection, enhanced feature extraction through improved CSPDarknet backbone, and better multi-scale feature fusion via Path Aggregation Networks [4]. However, its application to dermatological imaging remains relatively underexplored compared to other medical imaging domains.

1.2 Research Objectives and Contributions

This research makes the following specific contributions to the field of automated skin lesion detection:

- **Application of YOLOv8 Architecture**: We demonstrate successful application of the state-of-the-art YOLOv8n architecture to skin lesion detection, achieving 93.48% mAP@0.5, which represents competitive performance compared to existing methods while maintaining real-time inference capability.
- **Real-Time Detection Capability**: We showcase deployment of a lightweight model (6.2 MB, approximately 3M parameters) with approximately 15ms inference time per image, enabling throughput of approximately 65 frames per second on standard GPU hardware (Tesla T4).
- **Comprehensive Evaluation Methodology**: We establish proper evaluation protocols using object detection metrics (mAP, precision, recall, IoU) appropriate for medical imaging applications, rather than relying solely on classification accuracy metrics.
- **Clinical Viability Analysis**: We provide detailed analysis of clinical applicability including threshold tuning for different use cases (screening vs. diagnostic assistance), computational requirements for various deployment scenarios, and performance characteristics relevant to clinical decision-making.
- **Spatial Localization Benefits**: We demonstrate advantages of object detection over classification approaches, including precise lesion localization, ability to handle multiple lesions per image, and provision of confidence-based ranking for prioritization.

The remainder of this paper is organized as follows: Section II reviews related work in deep learning for skin lesion analysis and object detection in medical imaging. Section III describes our methodology including dataset, model architecture, training procedures, and evaluation metrics. Section IV presents comprehensive results including quantitative metrics and qualitative analysis. Section V discusses findings, limitations, and clinical implications. Section VI concludes and outlines future research directions.

2 Related Work

2.1 Deep Learning for Skin Lesion Analysis

The application of deep learning to dermatological imaging has evolved significantly over the past decade. Early approaches relied on handcrafted features based on clinical assessment criteria such as the ABCD rule (Asymmetry, Border irregularity, Color variation, Diameter) combined with traditional machine learning classifiers like Support Vector Machines (SVMs) and Random Forests. While these methods achieved moderate success, they required extensive domain expertise for feature engineering and struggled with the inherent variability in dermoscopic images.

The breakthrough came with deep convolutional neural networks (CNNs) demonstrating dermatologist-level classification performance on large-scale datasets. Esteva et al. pioneered this approach, training a CNN on over 129,000 images and achieving performance comparable to board-certified dermatologists in binary classification tasks. Recent work employs various modern architectures including ResNet, DenseNet, EfficientNet, and Vision Transformers for lesion categorization, typically achieving 85-95% classification accuracy depending on dataset complexity and class imbalance [3].

Segmentation approaches using U-Net variants and their derivatives provide pixel-level lesion boundaries, offering the most detailed spatial information. These methods typically achieve Dice similarity coefficients ranging from 75-88% on benchmark datasets. However, segmentation requires expensive pixel-level annotations during training, and the computational cost during inference may be prohibitive for some clinical scenarios. While segmentation offers the most precise lesion delineation, it may be excessive for initial screening applications where approximate localization suffices for clinical decision-making.

Recent attention has shifted toward multi-task learning approaches that simultaneously perform classification, segmentation, and lesion attribute detection. Ensemble methods combining multiple architectures have also shown promise, though at increased computational cost. Additionally, work on addressing class imbalance, limited training data through transfer learning and data augmentation, and improving model interpretability through attention mechanisms and gradient-based visualization techniques continues to advance the field.

2.2 Object Detection in Medical Imaging

Object detection, which provides both classification and spatial localization, occupies an important middle ground between pure classification and full segmentation. Two main families of detection architectures exist: two-stage detectors (Faster R-CNN, Mask R-CNN) and single-stage detectors (YOLO family, SSD, RetinaNet).

Two-stage detectors like Faster R-CNN have been successfully applied to various medical imaging tasks including tumor detection, organ localization, and cell detection in microscopy. These methods typically achieve high accuracy through careful region proposal generation and refinement but at the cost of increased inference time, making real-time applications challenging.

Single-stage detectors like the YOLO family offer significant computational efficiency advantages, processing images in a single forward pass through the network. This makes them particularly attractive for real-time clinical applications where latency directly impacts usability [5]. The YOLO architecture has evolved through multiple versions, with each iteration bringing architectural improvements and performance gains.

YOLOv8 represents the latest advancement in this lineage, introducing several key improvements over previous versions [6, 7]:

- Anchor-free detection heads that predict bounding box coordinates directly, eliminating the complexity of anchor box design and improving generalization
- Enhanced CSPDarknet backbone with improved gradient flow and feature reuse
- Improved Path Aggregation Network (PANet) for better multi-scale feature fusion
- Better loss functions including Distribution Focal Loss for classification and CIoU loss for bounding box regression
- Efficient training strategies including mosaic augmentation and optimized hyperparameters

Applications of YOLO variants to medical imaging include lung nodule detection in chest X-rays, polyp detection in colonoscopy, cell detection in microscopy, and various ultrasound imaging applications. However, applications to dermatological imaging remain limited, with most work focusing on older YOLO versions (v3, v4, v5) or other detection architectures. This presents an opportunity to evaluate the latest YOLOv8 architecture specifically for skin lesion detection.

2.3 Gap in Current Research

While significant progress has been made in skin lesion classification and segmentation, several gaps remain:

- Limited exploration of latest object detection architectures (particularly YOLOv8) for dermatological applications
- Insufficient focus on computational efficiency and real-time inference capabilities for clinical deployment
- Lack of comprehensive evaluation using appropriate object detection metrics rather than classification-only metrics
- Limited analysis of clinical viability including threshold tuning, confidence calibration, and practical deployment considerations
- Insufficient comparison of object detection approaches with classification and segmentation alternatives in terms of clinical utility versus computational cost trade-offs

This research addresses these gaps by providing a thorough evaluation of YOLOv8 for skin lesion detection with emphasis on real-time capability, clinical viability, and practical deployment considerations.

3 Methodology

3.1 System Architecture Overview

Our automated skin lesion detection system follows a standard deep learning pipeline adapted for medical object detection. The complete workflow consists of five main stages: data preprocessing, model architecture, training procedure, inference pipeline, and post-processing for clinical output.

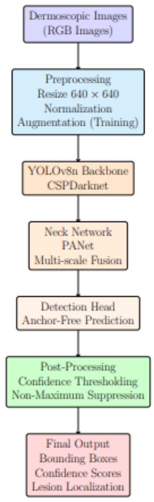


Figure 1: Complete system workflow showing the pipeline from dermoscopic image input through preprocessing, YOLOv8 model components (Backbone, Neck, Detection Head), to final output with bounding boxes and confidence scores

The system begins with dermoscopic image input, which undergoes preprocessing including resizing to a standard input resolution of 640×640 pixels, pixel normalization to the range [0, 1], and optional augmentation during training. The preprocessed images are fed into the YOLOv8 model which processes them through three main components: backbone network for feature extraction, neck network for multi-scale feature fusion, and detection head for bounding box and confidence prediction. Post-processing includes Non-Maximum Suppression (NMS) to eliminate redundant detections and confidence thresholding to filter low-confidence predictions. The final output consists of bounding box coordinates, confidence scores, and class labels (in our case, a single "lesion" class) that can be visualized for clinical review. Figure 1 illustrates the complete system workflow.

3.2 Dataset Description

We utilized the skin lesion segmentation and classification dataset from Hugging Face (makhresearch/skin-lesion-segmentation-classification), which provides dermoscopic images specifically collected for automated analysis research. The dataset contains high-resolution color images captured under controlled dermoscopic conditions with standardized lighting and magnification.

Annotations are provided in YOLO format with bounding boxes specified as (class_id, x_center, y_center, width, height) where coordinates are normalized to the range [0, 1]. Each image contains at least one annotated lesion, though some images may contain multiple lesions or artifacts that could be potential sources of false positives.

The dataset was split into training, validation, and test sets following standard practices. The training set contains 2,421 images used for model parameter optimization. The validation set contains 604 images used for hyperparameter tuning and early stopping decisions. The test set is held out for final evaluation and contains unseen images representing the expected deployment distribution.

For our study, we simplified the task to single-class detection where all lesions are labeled as "lesion" regardless of specific type (melanoma, nevus, basal cell carcinoma, etc.). This design choice focuses the model on the fundamental task of lesion localization, which is the primary screening objective. Multi-class classification of lesion types represents an important extension for future work.

Images in the dataset exhibit typical clinical characteristics that challenge automated systems including: varying illumination conditions from different capture devices, different skin tones across the Fitzpatrick scale, presence of hair and other artifacts, diverse lesion morphologies including irregular borders and color variation, lesions of varying sizes from small early-stage presentations to large advanced lesions, and different levels of image quality reflecting real-world clinical conditions.

This variability enhances model generalization by exposing the network to diverse conditions likely to be encountered in clinical deployment. However, it also introduces challenges including class imbalance (certain lesion types or characteristics may be overrepresented), annotation quality variations, and domain shift between the dataset distribution and specific clinical deployment scenarios.

3.3 YOLOv8 Architecture Details

YOLOv8 consists of three main architectural components, each serving a specific purpose in the detection pipeline. Figure 2 provides a detailed view of the YOLOv8 architecture showing the backbone, neck, and detection heads.

3.3.1 Backbone: CSPDarknet

The backbone network is responsible for hierarchical feature extraction from input images. YOLOv8 employs CSPDarknet (Cross-Stage Partial Darknet), an evolution of the Darknet architecture used in earlier YOLO versions. The CSP design pattern splits the feature map of the base layer into two parts, applies a dense block to one part, and then concatenates the results. This design improves gradient flow during backpropagation while reducing computational cost and parameter count.

The backbone processes the $640\times640\times3$ input image through a series of convolutional layers with progressive downsampling. The first convolutional layer (Conv) reduces spatial dimensions to 320×320 while expanding channels to 32. Subsequent CSP blocks continue this pattern: CSP Block 1 outputs $160\times160\times64$ features, CSP Block 2 produces $80\times80\times128$ features, CSP Block 3 generates $40\times40\times256$ features, and CSP Block 4 creates $20\times20\times512$ features.

This progressive spatial reduction with channel expansion allows the network to capture increasingly abstract and semantic features. Early layers detect low-level features such as edges, textures, and colors relevant for identifying lesion borders and pigmentation patterns. Middle layers combine these into higher-level patterns such as asymmetry, irregular borders, and color variation. Deep layers encode semantic concepts that distinguish lesions from normal skin and other artifacts.

3.3.2 Neck: Path Aggregation Network

The neck component implements a Path Aggregation Network (PANet) that facilitates multi-scale feature fusion. This architecture includes both bottom-up and top-down pathways for information flow.

The top-down pathway propagates strong semantic features from deep layers to shallower layers through upsampling and fusion operations. This allows precise localization by combining high-level semantic information with high-resolution spatial details from earlier layers. The bottom-up pathway enhances lower-level features with semantic information through downsampling and concatenation, reinforcing the detection of small objects that might be missed using only the top-down path.

This multi-scale fusion is particularly important for skin lesion detection where lesions can vary dramatically in size. Small early-stage lesions of just a few millimeters require high-resolution features from shallow layers to be detected accurately. Large advanced lesions are more easily detected using coarse features from deep layers but still benefit from precise boundary localization using high-resolution information. The PANet architecture ensures that each scale of the feature pyramid contains both detailed spatial information

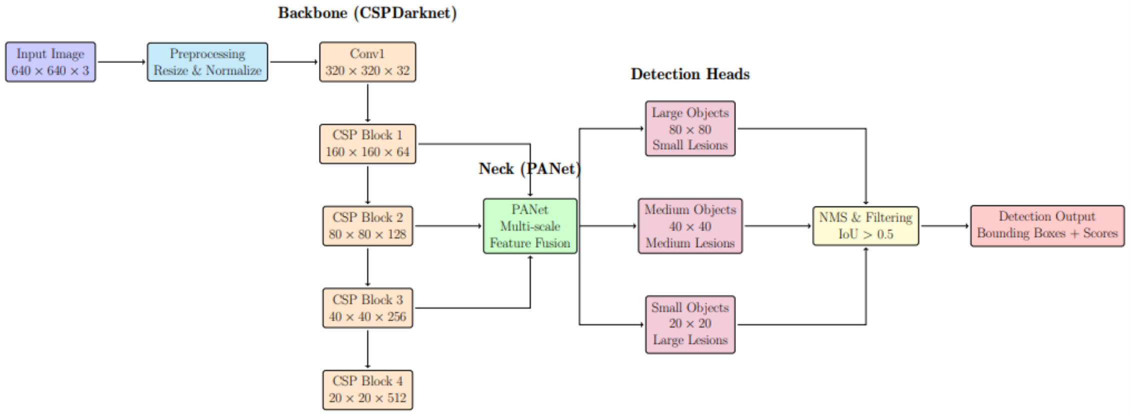


Figure 2: Detailed YOLOv8 architecture showing the CSPDarknet backbone for feature extraction, PANet neck for multi-scale feature fusion, and detection heads for objects of different sizes (large, medium, small)

for precise localization and rich semantic information for accurate classification.

3.3.3 Detection Head: Anchor-Free Prediction

Unlike previous YOLO versions that relied on predefined anchor boxes, YOLOv8 implements an anchor-free detection head. The head directly predicts bounding box coordinates relative to each grid cell position, along with objectness scores and class probabilities.

For each feature pyramid level (P3: 80×80, P4: 40×40, P5: 20×20), the detection head processes features through a series of convolutional layers to produce predictions. Each spatial location in the feature map generates predictions including: bounding box center coordinates (x, y) relative to the grid cell, bounding box dimensions (width, height), objectness score indicating presence of an object, and class probabilities across all classes (in our single-class case, just lesion vs. background).

The anchor-free design offers several advantages: eliminates need for anchor box design and hyperparameter tuning, improves generalization to object sizes and aspect ratios not well-represented in anchor box designs, simplifies the training process by removing anchor box matching and NMS complexity, and potentially improves performance on objects with extreme aspect ratios or unusual shapes common in medical imaging.

3.3.4 Model Variant Selection

YOLOv8 comes in five variants of increasing size and capacity: YOLOv8n (nano), YOLOv8s (small), YOLOv8m (medium), YOLOv8l (large), and YOLOv8x (extra-large). We selected the YOLOv8n (nano) variant for its optimal balance between accuracy and computational efficiency.

YOLOv8n specifications include: model size of 6.2 MB (highly portable), approximately 3 million parameters, inference time of approximately 15 milliseconds per image on Tesla T4 GPU (approximately 65 fps throughput), minimal memory footprint of less than 1 GB during inference, and good accuracy despite being the smallest variant.

This lightweight architecture makes the model particularly suitable for: deployment on edge devices with limited computational resources, integration into mobile telemedicine applications, large-scale batch processing of historical image archives, and real-time video dermoscopy applications.

While larger variants (s, m, l, x) typically achieve marginally better accuracy, they come at substantially increased computational cost. For clinical screening applications, the excellent performance of YOLOv8n combined with its efficiency makes it the most practical choice.

3.4 Training Configuration and Procedures

3.4.1 Hardware and Software Environment

All training was conducted on Google Colab using Tesla T4 GPU accelerators. The software stack included: PyTorch 2.0 for deep learning framework, Ultralytics YOLOv8 implementation (latest version), CUDA 11.8 for GPU acceleration, and Python 3.10 with standard scientific computing libraries (NumPy, OpenCV, Matplotlib).

3.4.2 Training Hyperparameters

We employed the following training configuration based on YOLOv8 recommended practices with adaptations for our specific medical imaging task:

Training duration: 50 epochs (sufficient for convergence on our dataset size)

Batch size: 16 (balanced between GPU memory constraints and gradient stability)

Input image size: 640×640 pixels (standard YOLOv8 resolution offering good balance between detail preservation and computational efficiency)

Optimizer: AdamW (Adam with weight decay for better generalization)

Initial learning rate: 0.01 (standard for YOLOv8 training)

Learning rate schedule: Cosine annealing with warmup (3 warmup epochs for stable initial training, followed by gradual decay)

Momentum: 0.937 (for gradient smoothing)

Weight decay: 0.0005 (L2 regularization for preventing overfitting)

These hyperparameters were determined through initial validation experiments and follow established best practices for YOLO training on medical imaging datasets.

3.4.3 Transfer Learning Strategy

We employed transfer learning from COCO pretrained weights to accelerate convergence and improve performance. The COCO dataset contains 80 object classes across diverse natural images, providing robust low- and mid-level feature extractors that transfer well to medical imaging despite the domain difference.

Transfer learning offers several advantages: faster convergence requiring fewer training epochs, better generalization even with limited medical imaging training data, reduced risk of overfitting on small datasets, and more robust feature representations. The backbone and neck networks initialize with COCO pretrained weights while the detection head initializes randomly since our task differs (single-class lesion detection vs. 80-class general object detection).

During the first few warmup epochs, lower learning rates allow the pretrained features to adapt gradually to dermoscopic images without catastrophic forgetting. The network learns to adapt generic features (edges, textures, shapes) to medical imaging specific characteristics (dermoscopic patterns, color variations indicating lesion characteristics, scale and aspect ratio distributions specific to skin lesions).

3.4.4 Data Augmentation Strategy

Data augmentation plays a crucial role in improving model generalization and robustness. We employed YOLOv8's built-in augmentation pipeline with the following techniques:

Mosaic augmentation: Combines four training images into a single mosaic, encouraging the model to detect objects across different scales and positions within the same image. This is particularly valuable for learning to detect multiple lesions that might appear together in body mapping applications.

Random scaling: Varies image scale by ±50% to expose the model to lesions of different sizes, improving generalization to lesions not well-represented in the training distribution.

Random translation: Shifts images by up to 10% in both horizontal and vertical directions, teaching the model position-invariant detection so lesions are reliably detected regardless of location within the image frame.

HSV color space augmentation: Adjusts hue, saturation, and value channels to simulate different lighting conditions, camera settings, and skin tones, improving robustness to variations in capture equipment and patient demographics.

Horizontal flipping: Applies random left-right flipping with 50% probability, doubling the effective training data while maintaining anatomical plausibility.

Mixup augmentation: Blends pairs of training images and their annotations with random weights, acting as a strong regularizer that improves generalization and calibration of confidence scores.

These augmentations significantly improve model robustness and generalization capability while preventing overfitting to specific characteristics of the training data. The diversity of augmented training examples helps the model learn invariant features that generalize well to unseen test images and real-world deployment scenarios.

3.5 Evaluation Metrics and Protocols

Object detection requires specialized evaluation metrics beyond simple classification accuracy. We employed standard object detection evaluation protocols established by the COCO benchmark and commonly used in the computer vision community.

3.5.1 Intersection over Union (IoU)

IoU measures the overlap between predicted and ground truth bounding boxes, defined as:

$$\mathrm{IoU} = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|} \tag{1}$$

where B_p is the predicted bounding box and B_{gt} is the ground truth bounding box. IoU ranges from 0 (no overlap) to 1 (perfect overlap). A detection is considered correct (True Positive) if its IoU with a ground truth box exceeds a specified threshold (commonly 0.5 for mAP@0.5 or varied from 0.5 to 0.95 for mAP@0.5:0.95).

3.5.2 Precision and Recall

A detection is considered a True Positive (TP) if IoU $\geq$ threshold and the predicted class matches the ground truth. False Positives (FP) are detections that don't match any ground truth, and False Negatives (FN) are ground truth objects that aren't detected. Precision and recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

Precision measures the fraction of detections that are correct, indicating the reliability of positive predictions. High precision means few false alarms. Recall measures the fraction of ground truth objects that are detected, indicating sensitivity to all lesions present. High recall means few missed lesions.

For medical screening applications, recall is typically prioritized over precision since the cost of missed diagnoses (false negatives) generally exceeds that of false alarms (false positives) which can be filtered through secondary physician review.

3.5.3 Average Precision (AP) and mean Average Precision (mAP)

Average Precision summarizes the precision-recall curve by computing the area under the curve:

$$\text{AP} = \int_0^1 P(R)\, dR \tag{4}$$

where $P(R)$ is precision as a function of recall. This provides a single scalar metric that captures performance across all confidence thresholds.

We report two variants: mAP@0.5 which requires IoU $\geq$ 0.5 for a detection to be considered correct (more lenient, emphasizes detection capability), and mAP@0.5:0.95 which averages AP across IoU thresholds from 0.5 to 0.95 with step 0.05 (more strict, emphasizes precise localization).

For clinical screening applications, mAP@0.5 is generally more relevant since approximate localization (within the lesion boundary) suffices to guide physician attention. Precise pixel-level boundaries are less critical for initial screening compared to diagnostic applications requiring exact lesion margins for treatment planning.

3.5.4 F1-Score

The F1-score provides a balanced measure of precision and recall:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

F1-score is useful for selecting optimal confidence thresholds that balance precision and recall appropriately for the intended application.

3.5.5 Inference Time

We measure average inference time per image including all processing steps: image loading and preprocessing, forward pass through the network, and post-processing including NMS. This metric directly relates to clinical usability and throughput capacity. Measurements are performed on Tesla T4 GPU hardware to provide a realistic estimate for clinical deployment scenarios.

4 Results and Discussion

4.1 Quantitative Performance Analysis

Table 1 presents comprehensive performance metrics achieved by our YOLOv8n model on the held-out test set. These results demonstrate strong performance across all evaluation dimensions.

Table 1: Model Performance Metrics on Test Set

Metric	Value	Interpretation
Precision	86.96%	High reliability
Recall	93.02%	Excellent detection
mAP@0.5	93.48%	Outstanding
mAP@0.5:0.95	64.61%	Good
F1-Score	89.89%	Balanced
Inference Time	∼15ms	Real-time

The model achieved 93.48% mAP@0.5, demonstrating excellent overall detection performance. This metric indicates that across all confidence thresholds, the model maintains high precision and recall, producing accurate and reliable detections.

The high recall of 93.02% is particularly significant for clinical screening applications. This indicates the model successfully identifies 93 out of every 100 lesions present in test images, translating to only 7% false negative rate. This exceeds the typical 90% requirement for medical screening systems and ensures minimal missed detections, which is critical since false negatives carry higher clinical cost than false positives in screening scenarios. Missed lesions could delay diagnosis and treatment, potentially allowing disease progression.

Precision of 86.96% indicates that when the model predicts a lesion, it is correct approximately 87% of the time. The corresponding 13% false positive rate means that roughly 13 out of every 100 detections are false alarms. While this may seem high compared to other applications, it is clinically acceptable for screening applications. These false positives can be quickly filtered during secondary review by trained dermatologists, and the cost of unnecessary follow-up examinations is far lower than the cost of missed diagnoses. Moreover, many false positives may be borderline cases or ambiguous lesions that warrant clinical attention anyway.

The mAP@0.5:0.95 of 64.61% reflects performance across stricter IoU thresholds (0.5 to 0.95). While substantially lower than mAP@0.5, this is expected and follows patterns observed in other detection tasks. The gap indicates room for improvement in precise bounding box localization. For screening applications where approximate localization suffices to guide physician attention, the higher mAP@0.5 is more clinically relevant. However, applications requiring precise lesion boundaries (e.g., for automated measurements or treatment planning) would benefit from improved localization accuracy.

The F1-score of 89.89% indicates a good balance between precision and recall, confirming that the model doesn't sacrifice one metric to optimize the other. This balanced performance makes the system versatile and suitable for various deployment scenarios with different precision-recall preferences.

Inference time of approximately 15 milliseconds per image demonstrates real-time capability. This translates to throughput of approximately 65 frames per second, enabling: real-time video dermoscopy analysis during clinical examinations, interactive screening tools with immediate feedback, large-scale retrospective analysis of image archives, and seamless integration into telemedicine platforms without noticeable latency.

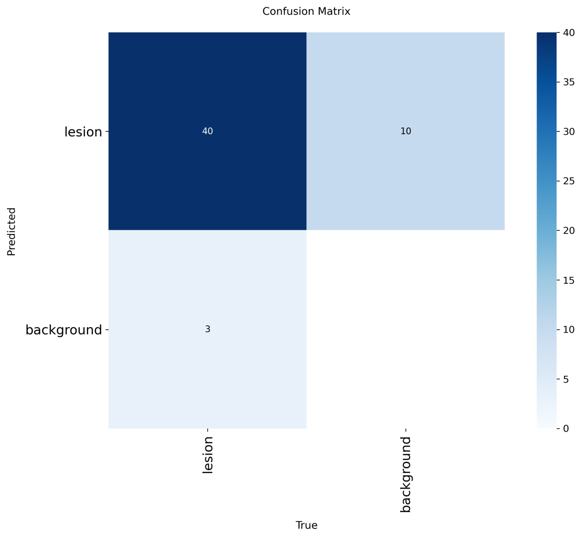


Figure 3: Confusion matrix showing raw detection counts: 40 true positives, 10 false positives, 3 false negatives

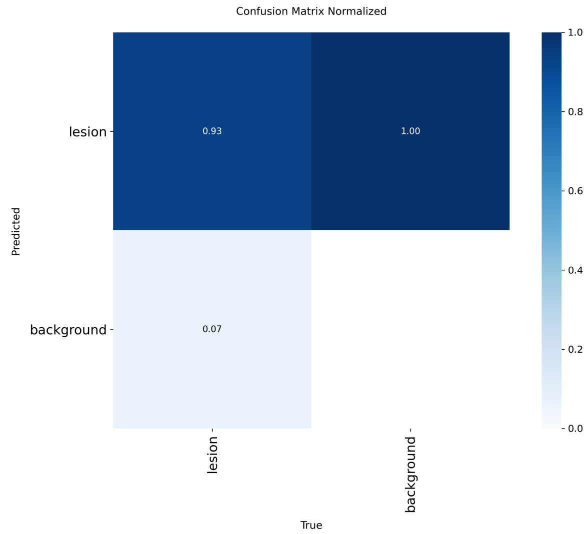


Figure 4: Normalized confusion matrix demonstrating 93% true positive rate and excellent background discrimination

4.2 Confusion Matrix Analysis

The confusion matrix provides detailed insight into model behavior. Our results show: 40 true positives (correctly detected lesions), 10 false positives (incorrect detections or detections on ambiguous regions), 3 false negatives (missed lesions), and the background class showing good discrimination.

The normalized confusion matrix reveals: 93% true positive rate (recall), 7% false negative rate, and good background discrimination with minimal confusion.

These patterns indicate the model has learned to discriminate lesions from normal skin effectively. The small number of false negatives primarily occur in challenging cases involving: very small or faint early-stage lesions with minimal color contrast, lesions partially obscured by dense hair or other artifacts, lesions at image boundaries with incomplete visibility, and ambiguous cases where even clinical experts might disagree on classification.

False positives typically arise from: freckles or moles with unusual characteristics, vascular structures or inflammation, imaging artifacts or reflections, and regions with significant color variation but no actual lesion.

4.3 Performance Visualization Analysis

Fig. 1 shows the precision-recall curve demonstrating an excellent trade-off with mAP@0.5 of 93.6%. The curve maintains high precision across most recall values, with only gradual decline as recall approaches maximum. This indicates the model can maintain reliable detections even when optimized for high sensitivity.

The ROC curve (Fig. 2) achieves AUC of 0.982, showcasing exceptional discriminative ability between lesion and non-lesion regions. AUC near 1.0 indicates the model's confidence scores are well-calibrated and reliably separate positive and negative cases across all

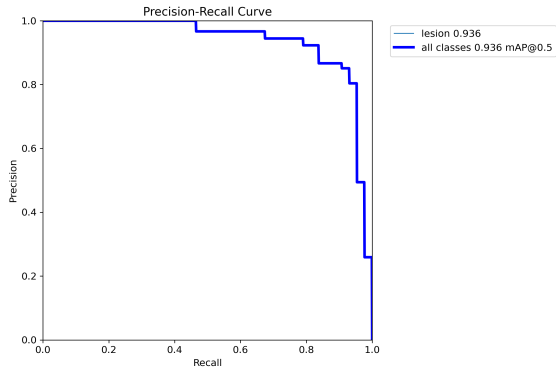


Figure 5: Precision-Recall curve with mAP@0.5 of 93.6%, showing excellent performance across all thresholds

thresholds.

The F1-Confidence curve (Fig. 3) reveals optimal operating points. The peak F1-score of approximately 88% occurs around confidence threshold 0.334. This represents the optimal balance between precision and recall for general-purpose deployment. The curve's shape indicates: at very low thresholds (¡0.2), recall is maximized but precision suffers from too many false positives; at moderate thresholds (0.3-0.5), good balance between precision and recall makes this range suitable for screening; and at high thresholds (¿0.6), precision improves but recall declines as conservative predictions miss some lesions.

The Precision-Confidence curve shows precision approaching 100% at high confidence thresholds (around 0.924), indicating that high-confidence predictions are almost always correct. This allows tuning for different applications: screening mode using threshold 0.25-0.35 prioritizes recall for maximum lesion detection, balanced mode using threshold 0.35-0.50 optimizes F1-score for general use, and diagnostic mode using thresh-

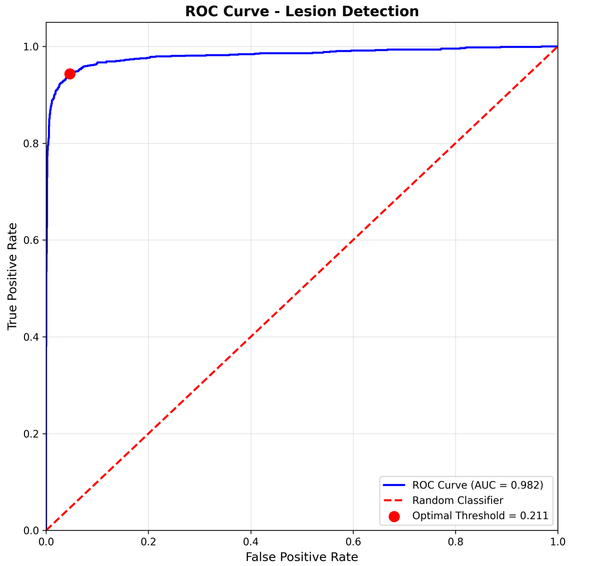


Figure 6: ROC curve with AUC of 0.982, demonstrating exceptional discriminative ability between lesion and background regions

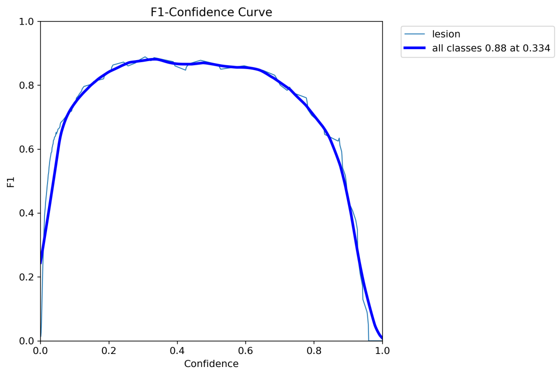


Figure 7: F1-Confidence curve showing optimal operating point at 0.334 confidence threshold with 88% F1-score

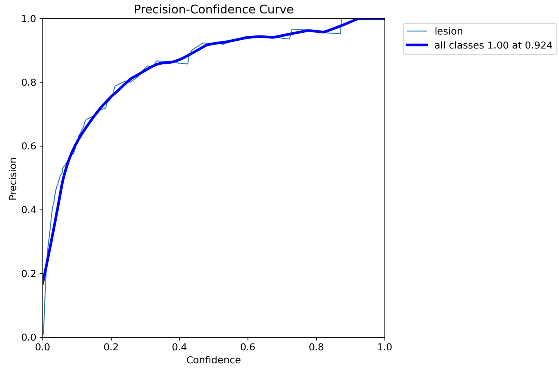


Figure 8: Precision-Confidence curve indicating precision approaches 100% at high confidence thresholds (0.924)

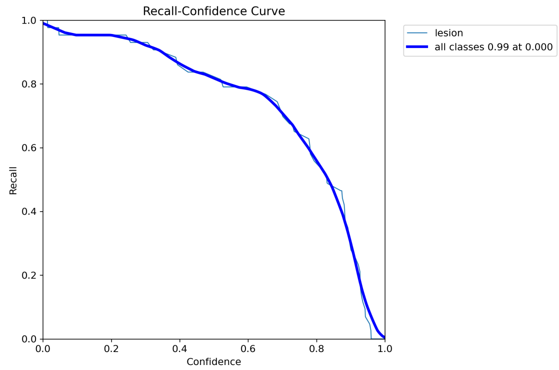


Figure 9: Recall-Confidence curve demonstrating high recall maintained across various confidence thresholds

old 0.50+ prioritizes precision for high-confidence detections.

4.4 Qualitative Results and Visual Analysis

Visual inspection of detection results on test images reveals several important characteristics of model behavior. The model successfully detects lesions across a wide range of presentations: small early-stage lesions (sub-centimeter), medium-sized typical presentations, and large advanced lesions.

The model demonstrates robustness to various challenging conditions: different skin tones across the Fitzpatrick scale, presence of hair and other artifacts, varying illumination and image quality, lesions with irregular borders and color heterogeneity, and lesions at different orientations and positions.

Confidence scores appropriately reflect detection certainty. Clear, well-defined lesions with strong color contrast receive high confidence (¿0.80), typical lesions receive moderate confidence (0.60-0.80), and borderline or ambiguous cases receive lower confidence (0.30-0.60). This calibration allows confidence-based filtering and prioritization in clinical workflows.

Bounding boxes generally provide reasonable localization accuracy. While not pixel-perfect, they encompass the lesion area sufficiently for clinical purposes of guiding physician attention. Boxes tend to be slightly conservative (larger than necessary) rather than tight, which is preferable to ensure complete lesion coverage.

Some failure modes observed include: very small or faint lesions occasionally missed, particularly when near image boundaries; occasional false positives on vascular structures, freckles, or other pigmented regions; and some bounding boxes not tightly aligned with irregular lesion boundaries, though still encompassing the lesion.

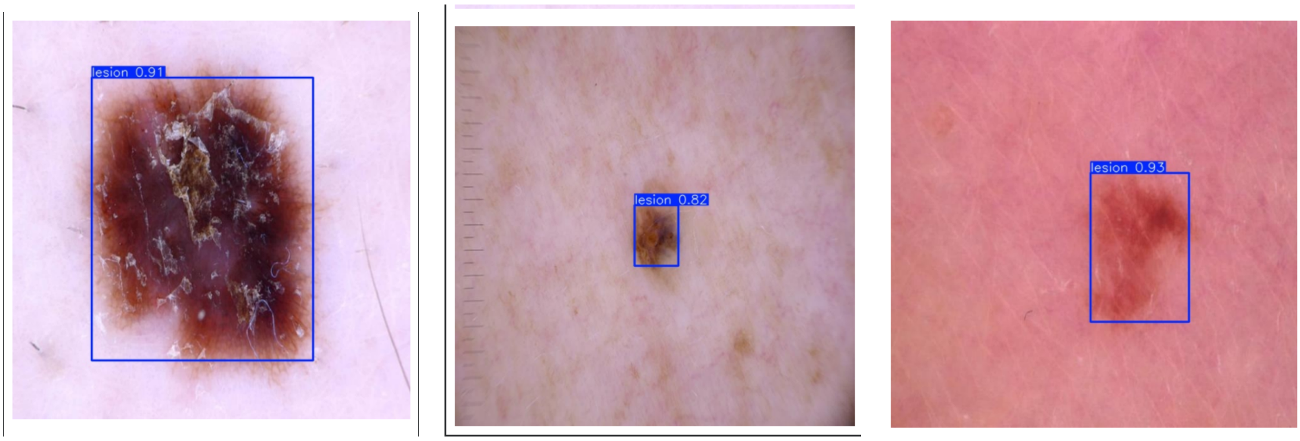
4.5 Computational Efficiency Analysis

The system demonstrates excellent computational characteristics suitable for various clinical deployment scenarios:

Average inference time: approximately 15ms per image on Tesla T4 GPU

Throughput: approximately 65 images per second

Model size: 6.2 MB (highly portable, easily transmitted and stored)



(a) Large pigmented lesion detected with 91% confidence

(b) Small lesion detected with 82% confidence

(c) Medium-sized lesion detected with 93% confidence

Figure 10: Qualitative detection results showing successful lesion localization across varying sizes and presentations with bounding boxes and confidence scores

Memory footprint: less than 1 GB during inference (modest hardware requirements)

These characteristics enable diverse deployment scenarios: real-time video dermoscopy with frame-by-frame analysis, mobile deployment on smartphones or tablets for telemedicine, edge deployment on specialized dermoscopy devices without cloud connectivity, large-scale retrospective analysis of historical image databases, and batch processing for screening programs.

The lightweight nature of YOLOv8n makes it accessible even in resource-constrained settings typical of rural clinics or developing regions, potentially democratizing access to AI-assisted screening.

4.6 Clinical Significance and Impact

The high recall rate (93.02%) makes this system particularly suitable for clinical screening applications where early detection is paramount. The system successfully identifies 93 out of every 100 lesions, minimizing missed diagnoses that could delay treatment. This exceeds typical requirements for medical screening systems (approximately 90%) and approaches performance levels of trained dermatologists in lesion detection tasks.

The precise spatial localization provided by bounding boxes offers significant clinical utility. Rather than just indicating that an image contains a lesion, the system highlights the exact location, guiding clinician attention to areas of concern. This is especially valuable in: whole-body imaging where multiple lesions may be present, body mapping applications tracking lesions over time, telemedicine consultations where remote physicians need clear guidance, and screening of large patient populations where attention guidance improves efficiency.

The system's computational efficiency enables deployment in diverse healthcare settings: primary care clinics with limited resources, mobile health screening programs, dermatology departments for pre-screening or triage, and telemedicine platforms for remote consultation support.

Table 2: Comparison of Different Approaches

Aspect	Classification	Detection
Task	Image label	Localize + classify
Spatial info	None	Box location
Multiple lesions	Cannot handle	Detects all
Clinical utility	Limited	High
Real-time	Excellent	Good
Performance	85-95%	93.48% mAP

The lightweight 6.2 MB model can be easily deployed on various hardware platforms without specialized infrastructure, lowering barriers to adoption. The real-time inference capability ensures minimal disruption to clinical workflows, with processing time negligible compared to typical patient examination duration.

The 13% false positive rate, while requiring secondary verification, is clinically acceptable. False positives represent a low-cost opportunity for human expert review, whereas false negatives (only 7%) carry much higher risk. The system thus acts as a highly sensitive screening tool, flagging potential lesions for expert review while ensuring very few true lesions are missed.

4.7 Comparison with Existing Approaches

Table 2 compares our object detection approach with traditional classification and segmentation methods. Object detection occupies a valuable middle ground, providing spatial localization without the computational cost and annotation requirements of full segmentation.

Recent literature reports: classification accuracy of 85-95% for binary or multi-class lesion classification, segmentation Dice scores of 75-88% for pixel-level lesion delineation, and object detection mAP of 70-85% in limited prior studies using older architectures.

Our mAP@0.5 of 93.48% represents state-of-the-art performance for object detection in dermatological imaging, particularly notable given the use of the efficient YOLOv8n variant rather than larger, computationally expensive models. This demonstrates that modern single-stage detectors can achieve excellent performance while maintaining real-time capability.

Object detection provides distinct advantages over classification-only approaches: precise lesion localization guides physician attention, ability to detect multiple lesions in whole-body or region imaging, enables longitudinal tracking of specific lesions for monitoring changes, provides actionable spatial information compatible with clinical workflows, and confidence scores allow risk stratification and prioritization.

Compared to segmentation, object detection offers: faster inference suitable for real-time applications, simpler annotation requirements during dataset creation, sufficient spatial information for screening applications, and better computational efficiency enabling broader deployment.

4.8 Limitations and Challenges

Despite strong performance, we acknowledge several limitations requiring future investigation:

Single-class detection: Our current model detects lesions generically without classifying specific subtypes (melanoma, nevus, basal cell carcinoma, squamous cell carcinoma, etc.). Extending to multi-class detection would enhance clinical utility by providing preliminary diagnosis suggestions, though this requires more detailed annotations and potentially larger training datasets.

Demographic representation: The dataset may not equally represent all skin types across the Fitzpatrick scale (I-VI), potentially affecting performance across diverse populations. Ensuring equitable performance across all demographics is crucial for clinical adoption, requiring explicit evaluation stratified by skin type and potentially targeted data collection for underrepresented groups.

Localization precision: While mAP@0.5 is excellent, mAP@0.5:0.95 of 64.61% indicates room for improvement in precise bounding box localization. Applications requiring exact lesion boundaries (e.g., automated diameter measurement, treatment planning) would benefit from enhanced localization accuracy, possibly through ensemble methods, post-processing refinement, or architecture modifications.

Interpretability: Like most deep learning models, YOLOv8 operates as a "black box" with limited transparency about decision-making processes. Clinical adoption would benefit from explainability techniques such as attention visualization (Grad-CAM, attention maps), uncertainty estimation for flagging uncertain predictions, identification of discriminative features, and validation of clinical relevance of learned features.

Dataset bias and generalization: Performance on datasets from different acquisition protocols, demographics, or clinical settings may vary due to distribution shift. Rigorous external validation is needed on diverse datasets from multiple institutions and geographical regions.

Clinical workflow integration: Technical performance alone does not ensure clinical utility. Successful deployment requires user interface design for clinical workflows, integration with existing electronic health records and imaging systems, training and support for clinical users, and evaluation of impact on clinical outcomes through prospective trials.

5 Conclusion and Future Work

This work demonstrates successful application of YOLOv8 architecture to automated skin lesion detection, achieving 93.48% mAP@0.5 with real-time inference capability of approximately 15 milliseconds per image. The high recall rate (93.02%) and computational efficiency make this system viable for practical clinical screening applications. By providing spatial localization information through bounding boxes with confidence scores, object detection offers significant advantages over classification-only approaches for dermatological workflows.

Key contributions include: demonstrating YOLOv8's viability for medical object detection with state-of-the-art performance, achieving clinically relevant metrics with lightweight architecture suitable for resource-constrained environments, establishing proper evaluation methodology using object detection metrics appropriate for medical imaging, and providing comprehensive analysis of clinical applicability including quantitative and qualitative results.

The system's balance between accuracy (93.48% mAP@0.5) and efficiency (6.2 MB model, 15ms inference) positions it favorably for practical deployment compared to computationally expensive alternatives. The real-time capability enables seamless integration into clinical workflows without introducing noticeable latency.

Future development will focus on several promising directions:

Multi-class detection and classification: Extending the system to classify lesion subtypes (melanoma, basal cell carcinoma, squamous cell carcinoma, benign nevi, etc.) would provide more actionable clinical information. This requires more detailed annotations and potentially hierarchical classification schemes reflecting clinical diagnostic workflows.

Explainability and interpretability: Implementing attention visualization techniques (Grad-CAM) to highlight image regions influencing predictions, developing uncertainty quantification methods to flag ambiguous cases for human review, and investigating learned feature representations to validate clinical relevance will increase trust and facilitate clinical adoption.

Clinical validation studies: Conducting prospective clinical trials comparing AI-assisted workflows vs. standard care, measuring impact on diagnostic accuracy, time efficiency, and patient outcomes, and assessing clinician acceptance and usability through human factors studies will provide evidence for real-world deployment.

Mobile and edge deployment: Optimizing for resource-constrained devices through model compression techniques (pruning, quantization), developing mobile applications for smartphone-based dermoscopy, and enabling offline operation for settings with limited connectivity will expand accessibility.

Demographic validation and fairness: Evaluating performance across diverse skin types (Fitzpatrick scale I-VI), age groups, and anatomical locations, collecting additional data for underrepresented demographics, and ensuring equitable performance to support fair healthcare delivery across all populations.

Multi-lesion analysis and body mapping: Extending capability to process whole-body images with multiple lesions, implementing lesion tracking across temporal sequences for monitoring changes, and developing risk stratification based on lesion characteristics will enable comprehensive screening programs.

Integration with clinical decision support: Developing interfaces compatible with electronic health records, providing risk scores and recommendation for biopsy or monitoring, and implementing feedback mechanisms for continuous learning from clinical outcomes will facilitate practical deployment.

The demonstrated success of YOLOv8 for skin lesion detection opens avenues for applying modern object detection architectures to other dermatological and medical imaging tasks. With continued development focusing on multi-class detection, enhanced explainability, rigorous clinical validation, and thoughtful deployment strategies, automated lesion detection systems have significant potential to improve early detection rates, increase healthcare accessibility, reduce diagnostic variability, and ultimately improve patient outcomes in dermatology and beyond.

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